

Chapter

8

Electrostatic Force

ELECTROSTATICS FORCE: The electrostatic force is an attractive or repulsive force between particles that are caused due to their electric charges.

Few important points are:

- **Amber Experiment:** Early Greek philosophers studied electrostatics through an experiment involving amber. When amber was rubbed with silk or wool, it attracted dust particles when brought close. The force produced in this process is known as electrostatic force.
- **Attraction Between Amber and Dust:** Dust particles would jump to stick to the rubbed amber, demonstrating the attractive force known as electrostatic force.
- **Electrostatic vs. Gravitational Force:** The electrostatic force between charges is similar to the gravitational force. However, while charged objects can (i) attract or (ii) repel each other via electrostatic force, gravitational force only attracts.
- **Nature of Charged Objects:** Two charged objects experience electrostatic force, which can either draw them closer or push them apart.
- **Gravitational Attraction:** Unlike electrostatic force, gravitational force is always attractive, pulling objects toward each other.
- **Short Range vs. Strong Force:** Electrostatic forces are short-ranged, acting over smaller distances compared to gravitational force. However, they are significantly stronger.
- **Interaction Based on Charge Nature:** Objects with the same charge repel each other, while objects with opposite charges are attracted through electrostatic force.

COLOUMBS LAW:

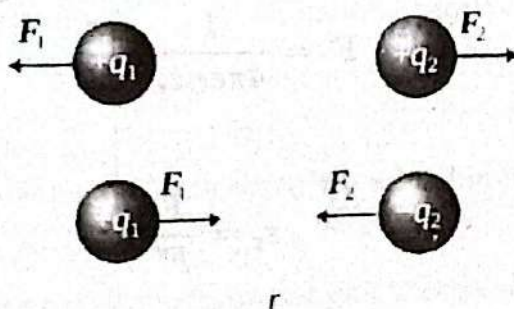
Introduction

The magnitude of the force of attraction or repulsion between two electric charges at rest was studied by French Scientist Charles Coulomb. He formulated a law, known as "COULOMB'S LAW".

Statement

According to Coulomb's law:

- The electrostatic force of attraction or repulsion between two point charges is directly proportional to the product of the charges.
- The electrostatic force of attraction or repulsion between two point charges is inversely proportional to the square of distance between them.



$$|F_1| = |F_2| = k_e \frac{|q_1 \times q_2|}{r^2}$$

MATHEMATICAL REPRESENTATION OF COULOMB'S LAW

Consider two point charges q_1 and q_2 placed at a distance of r from each other. Let the Electrostatic force between them is F .
According to the first part of the law

$$F \propto q_1 q_2$$

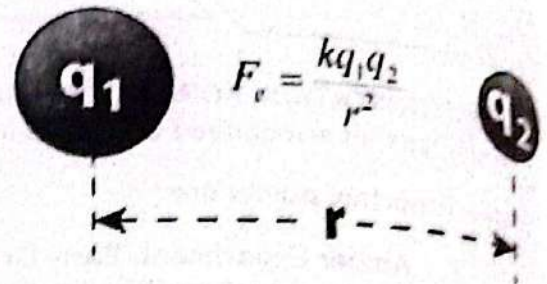
According to the second part of the law

$$F \propto \frac{1}{r^2}$$

Combining above statements

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = K \frac{q_1 q_2}{r^2}$$



Where k is the constant of proportionality.

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2}$$

Value and Unit of K

Value of K is equal to $\frac{1}{4\pi\epsilon_0}$

Where ϵ_0 is permittivity of free space. Its value is $8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$.

Thus in S.I. system numerical value of K is $8.98755 \times 10^9 \text{ Nm}^2\text{C}^{-2}$.

Force In The Presence Of Dielectric Medium

If the space between the charges is filled with a non-conducting medium or Insulator called a "Dielectric", it is found that the dielectric reduces the electrostatic force compared to free space by a factor known as DIELECTRIC CONSTANT (denoted by ϵ_r). This factor is also known as RELATIVE PERMITTIVITY and has different values for different dielectric materials. In the presence of a dielectric between two charges the Coulomb's law is expressed as:

$$F' = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q_1 q_2}{r^2}$$

Dividing F by F' so,

$$\epsilon_r = \frac{F}{F'}$$

Vector Form of Coulomb's Law

The magnitude as well as the direction of electrostatic force can be expressed by using Coulomb's law by vector equation

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

Where $\vec{F}_{12}$ is the force exerted by q_1 on q_2 and $\hat{r}_{12}$ is the unit vector along the line joining the two charges from q_1 to q_2 .

Limitations of Coulomb's Law

- We can use the formula if the charges are static (In rest position)
- The formula is easy to use while dealing with charges of regular and smooth shape, and it becomes too complex to deal with charges having irregular shapes

Definition of Coulombs

Coulomb is the S.I unit of charge. A unit coulomb is defined as the amount of charge that flow through a given cross section of a wire in one second, if there is a steady current of 1 ampere in the wire.

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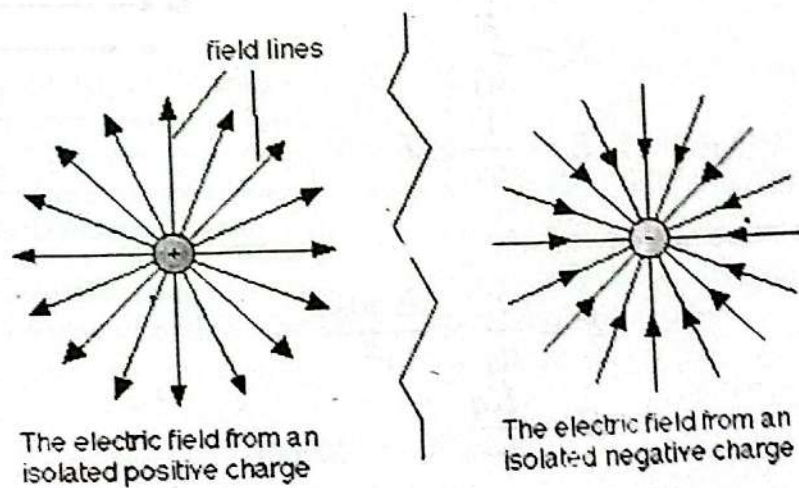
Electric field

ELECTRIC FIELD:

Space or region around an electric charge or a charged body within which another charge experiences some electrostatic force of attraction or repulsion is called Electric Field.

OR

Space or region around an electric charge or a charged body where its effect is felt.



ELECTRIC FIELD INTENSITY

Definition

Electric Intensity at a point is defined as the force experienced per unit positive test charge (q_0) at a point placed in the electric field.

OR

It may also be also defined as the electrostatic force per unit charge which the field exerts at a point.

Mathematically

$$E = \frac{F}{q_0}$$

Unit

The unit of E is N/C (or Newton/Coulomb).

Factors Affecting ForceThe force experienced by a charge $+q_0$ in an electric field depends upon.

- Magnitude of test charge (q_0)
- Strength of electric field (E)

Nature and Direction Of Electric Field Intensity

- It is a vector quantity.
- It is directed away from positive charge and directed toward negative charge.
- It has the same direction as that of force.

ELECTRIC FIELD INTENSITY NEAR AN ISOLATED POINT CHARGEConsider a point charge q called SOURCE CHARGE placed at a point 'O' in space. To find its intensity at a point 'p' at a distance 'r' from the point charge we place a test charge ' q_0 '.The force experienced by the test charge q' will be

$$F = Eq_0$$

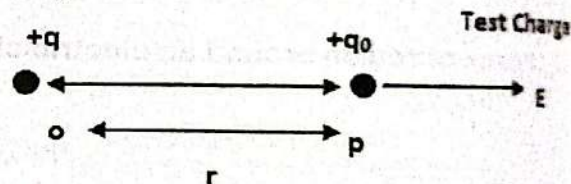
$$F = \frac{Kqq_0}{r^2}$$

We know that:

$$E = \frac{F}{q_0}$$

$$E = \frac{1}{q_0} \times F$$

Point Charge



Putting the value of 'F' we get

$$E = \frac{1}{q_0} \times \frac{Kqq_0}{r^2}$$

$$E = \frac{Kq}{r^2}$$

$$\text{But } K = \frac{1}{4\pi\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

This shows that the electric intensity due to a point charge is directly proportional to the magnitude of charge q and inversely proportional to the square of distance

If there are n number of charged particles $q_1, q_2, q_3, \dots, q_n$ then the net electric force of these charges at a point. According to superposition principle,

$$F = F_1 + F_2 + F_3 + \dots$$

$$E = E_1 + E_2 + E_3 + \dots$$

and similarly:

Therefore, the electric fields of these charges experienced by the test charge q_0 is vector sum of electric field of individual charge.

Effect of Dielectric Medium

If there is a medium of dielectric constant (ϵ_r) between the source charge and the field charge, intensity at a point will decrease ϵ_r time's i.e.

$$E = \frac{1}{4\pi\epsilon_0\epsilon_r} \times \frac{q}{r^2}$$

Vectorial Form

$$\vec{E} = \frac{1}{4\pi\epsilon_0\epsilon_r} \times \frac{q}{r^2} \hat{r}$$

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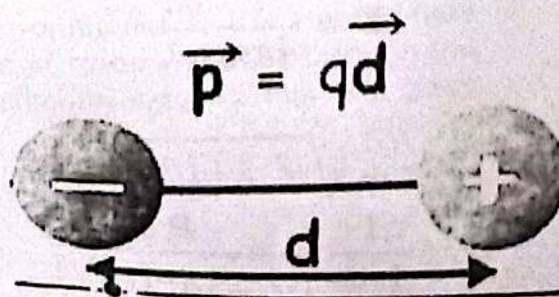
Electric Dipoles

ELECTRIC DIPOLE

A dipole is a separation of opposite electrical charges and it is quantified by an electric dipole moment. The **electric dipole moment** associated with two equal charges of opposite polarity separated by a distance, d is defined as the vector quantity having a magnitude equal to the product of the charge and the distance between the charges and having a direction from the negative to the positive charge along the line between the charges.

It is a useful concept in dielectrics and other applications in solid and liquid materials. These applications involve the energy of a dipole and the electric field of a dipole.

ELECTRIC DIPOLE



Physics XI

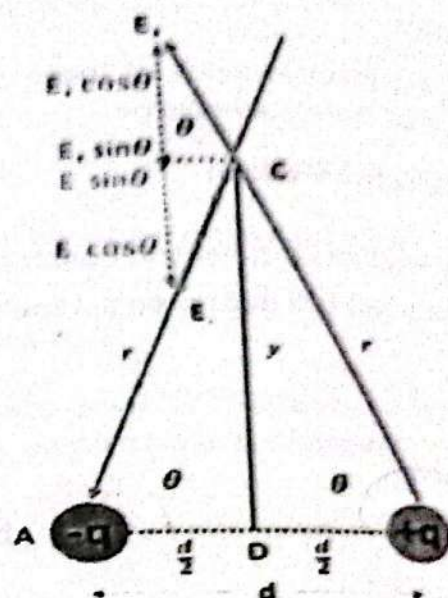
ELECTRIC FIELD AT A POINT DUE TO TWO CHARGES:

An electric dipole consists of equal and opposite point charges separated by a distance. To calculate the electric field at point C, consider two charges " $-q$ " and " $+q$ " at a distance " d ".

The charge " $+q$ " sets up field E_+ and the charge " $-q$ " produce the field E_- . Resolving the field vectors into the components. From figure, it is clear that the vertical components " $E_+ \sin \theta$ " and " $E_- \sin \theta$ " cancel each other. The net electric field is due to the vector sum of the horizontal components. In order to derive the Electric field first we have to resolve the E_+ and E_- into its horizontal and vertical component as shown in the diagram.

Components of $E_+ = E_+ \cos \theta$ and $E_+ \sin \theta$

Components of $E_- = E_- \cos \theta$ and $E_- \sin \theta$



As from the diagram we can say that the net Electric field is due to horizontal component of vector therefore

$$E = E_+ \cos \theta + E_- \cos \theta$$

As both charges are equal.

$$E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$E = 2 \times \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \cos \theta$$

From figure, In triangle ADC,

$$\cos \theta = \frac{d/2}{r} = \frac{d}{2r}$$

Putting the value of $\cos \theta$, we get.

$$E = 2 \times \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) \frac{d}{2r}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{r^3}$$

From figure, In triangle ΔADC ,

$$r = \sqrt{y^2 + (d/2)^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{(y^2 + (d/2)^2)^{3/2}}$$

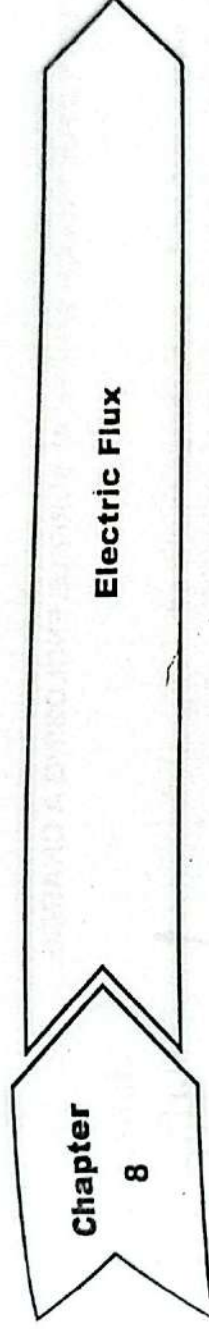
where the product of the "q" and "d" is known as the electric dipole moment "P" of the dipole. The electric dipole moment is a vector quantity and it is always directed from the negative to the positive charge.

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3 (1 + (d/2y)^2)^{3/2}}$$

As $d \ll y$, therefore approximate electric field intensity of the dipole can be calculated by neglecting the term $(d/2y)^2$.

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3}$$

The direction of electric dipole moment is always opposite to direction of net Electric field. i.e. Away from negative and towards positive.



ELECTRIC FLUX

In common language flux refers to the flow or stream of anything from one point to another point. In the similar way electric flux is the total number of lines of force passing through a surface.

Physical Meaning of Electric Flux

The total number of lines of force passing through the unit area of a surface held perpendicularly.

Mathematical Meaning of Electric Flux

The dot product of electric field intensity (E) and the vector area ($\vec{A}$) is called electric flux.

$$\Delta\phi_e = \vec{E} \cdot \vec{\Delta A}$$

$$\Delta\phi_e = E \Delta A \cos \theta$$

Where θ is the angle between E and ΔA

Maximum Flux

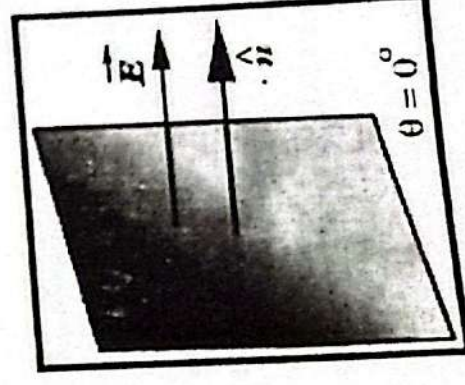
If the surface is placed perpendicular to the electric field then maximum electric lines of force will pass through the surface. Consequently maximum electric flux will pass through the surface.

$$\phi_e = \vec{E} \cdot \vec{\Delta A}$$

$$\phi_e = E \Delta A \cos 0^\circ$$

$$\phi_e = E \Delta A (1)$$

$$\phi_e = E \Delta A$$



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Zero Flux

If the surface is placed parallel to the electric field then no electric lines of force will pass through the surface. Consequently no electric flux will pass through the surface.

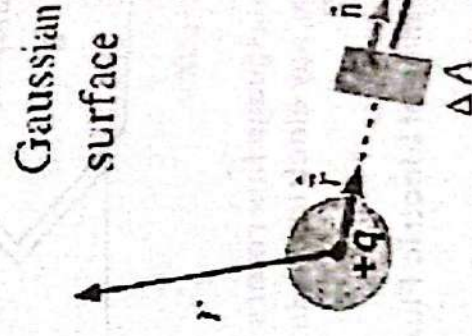
$$\begin{aligned}\phi_e &= \vec{E} \cdot \vec{\Delta A} \\ \phi_e &= E \Delta A \cos 90^\circ \\ \phi_e &= E \Delta A (0) \\ \phi_e &= 0\end{aligned}$$

Unit and Nature Of Flux

- Its unit is $\text{N.m}^2/\text{coulomb}$ or V.m
- Flux is a scalar quantity.

ELECTRIC FLUX THROUGH SPHERICAL SURFACE ENCLOSING A CHARGE:

Consider a small positive point charge $+q$ placed at the center of a closed sphere of radius " r ". The general relation for flux is not applicable in this situation because the direction of electric intensity varies point to point over the surface of sphere. In order to overcome this problem the sphere is divided into a number of small and equal pieces each of area ΔA . The direction of electric field in each segment of sphere is the same i.e. outward normal.



Now we will determine the flux through each segment
Electric flux through the first segment

$$\begin{aligned}\Delta\phi_1 &= E \Delta A \cos \theta \\ \text{but } \theta &= 0 \\ \Delta\phi_1 &= E \Delta A \cos 0 \\ \Delta\phi_1 &= E \Delta A (1) \\ \Delta\phi_1 &= E \Delta A\end{aligned}$$

Electric flux through the second segment

$$\Delta\phi_2 = E \Delta A$$

Electric flux through other segments

$$\Delta\phi_3 = E\Delta A$$

$$\Delta\phi_n = E\Delta A$$

Being a scalar quantity, the total flux through the sphere will be equal to the algebraic sum of all these flux i.e.

$$\phi = \sum_{i=1}^{i=n} \phi_i$$

$$\phi = \sum_{i=1}^{i=n} (E\Delta A)_i$$

$$\phi = E \sum_{i=1}^{i=n} (\Delta A)_i$$

But $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \sum_{i=1}^{i=n} (\Delta A)_i$$

$$\sum_{i=1}^{i=n} (\Delta A)_i = 4\pi r^2$$

As the area of sphere is

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} 4\pi r^2$$

$$\phi = \frac{q}{\epsilon_0}$$

$$\frac{1}{\epsilon_0}$$

This expression shows that the total flux through the sphere is $\frac{1}{\epsilon_0}$ times the charge enclosed (q) in the sphere. This is Gauss's Law

CONCLUSION: The total flux through closed sphere is independent of the radius of sphere.

ELECTRIC POTENTIAL:

Electric potential energy and electric potential are measures of how much energy a charge has because of its position in an electric field.

In short:

- Electric potential energy: Energy stored by a charge in an electric field.
- Electric potential: Energy per unit charge.
- Moving a charge against the electric field requires energy.
- Electric potential energy is like the effort you put into lifting an object.
- Electric potential is like the "height" in the electric field.

Definition

Potential difference between two points in an electric field is defined as

The work done in moving a test charge from one point to another against electric field divided by the magnitude of test charge

Mathematically

$$V = \frac{W}{q_0} = \frac{U}{q_0}$$

NATURE AND UNIT

It is a scalar quantity.

Its unit is Joule/Coulomb (or Volt)

Electric Potential at a Point

Think of electric potential like a special number we can find. Imagine there's a charge, let's call it "q". and it makes an electric field around it. Now, imagine we have a tiny test charge "q₀" placed some distance away from this charge.

The work it takes to move the test charge "q₀" is related to a number we call electric potential, written as "V."

As we know that

$$V = \frac{W}{q_0} = \frac{F r}{q_0}$$

$$V = \frac{E q_0 r}{q_0}$$

$$\text{where } F = E q_0$$

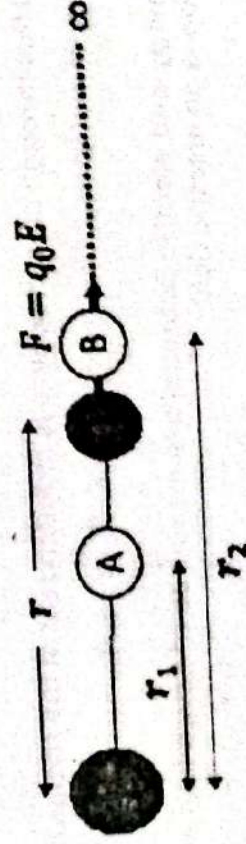
$$V = E r$$

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

By putting the value of

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

Figure below shows the potential at the two different positions in the field of charge "q"



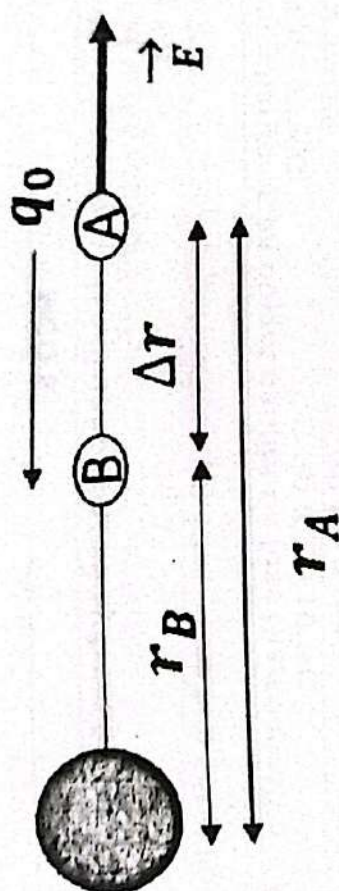
The potential difference can be written as:

$$V_2 - V_1 = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$$

where V₁ and V₂ are electric potentials at a distance r₁ and r₂

RELATION BETWEEN ELECTRIC INTENSITY AND POTENTIAL DIFFERENCE:

Suppose a positive test charge q_0 from A to B against the Electric Field. The work done in moving test charge q_0 through a displacement against the field is



$$\Delta V = E \cdot \Delta r = E \Delta r \cos \theta$$

$$\Delta V = -E \Delta r \quad [\because \text{Displacement is against electric field}]$$

$$E = -\frac{\Delta V}{\Delta r}$$

$$\frac{\Delta V}{\Delta r}$$

But $\frac{\Delta V}{\Delta r}$ is called potential gradient. Hence E is equal to potential gradient.

Definition of Potential Gradient

Negative rate of change of potential with respect to displacement.

Electron Volt

It is the unit of energy. It is defined as the energy acquired by an electron in falling through a potential difference of 1 volt.

$$1 \text{ electron volt} = (1.6 \times 10^{-19} \text{ Coulombs}) \times (1 \text{ volt}) = 1.6 \times 10^{-19} \text{ Joules}$$

MCQ'S

Ans	
c	1. A $2\mu\text{C}$ point charge is located a distance "d" away from $6\mu\text{C}$ point charge, what is the ratio of F_{12}/F_{21} ? (a) $1/3$ (b) 3 (c) 1 (d) 12
a	2. The minimum charge on an object cannot be less than. (a) $1.6 \times 10^{-19}\text{C}$ (b) $3.2 \times 10^{-19}\text{C}$ (c) 9.1×10^0 (d) No definite value exist
b	3. Two charges are placed at a certain distance. If the magnitude of each charge is doubled the force will become (a) $1/4$ th of its original value (b) 4 times of its original value (c) $1/8$ th of its original value (d) 8 times of its original value
c	4. Which of the following can be deflected while moving in the electric field? (a) neutron (b) photon (c) electron (d) both a and b
b	5. The flux through a flat surface of area "A" in a uniform electric field "E" is maximum when the surface area is: (a) Parallel to E (b) perpendicular to E (c) placed 45° to E (d) placed 60° to E
c	6. The product of charge "q" and small separation "d" between two charges of same magnitude and opposite in nature is known as: (a) Electric dipole (b) Moment arm (c) Electric dipole moment (d) Flux of electric field
b	7. 12 J of work is to be done against an existence electric field to take a charge of 0.01 C from one-point A to another point B. The potential difference between B and A is (a) 120 V (b) 1200 V (c) 1.2 V (d) 12 V
b	8. The force between two charges placed in air is F. If air is replaced by a medium of relative permittivity ϵ_r then force is reduced to: (a) $F\epsilon_r$ (b) F / ϵ_r (c) ϵ_r / F (d) $\epsilon_o \epsilon_r$
c	9. The negative gradient of the potential is: (a) potential energy (b) voltage (c) electric field intensity (d) electric flux
b	10. The electric flux through a plane area will be half of its maximum value when area is held at angle of with electric field (a) 30° (b) 60° (c) 45° (d) 90°

***** Exams Practice MCQs *****

1. The electric flux through a surface will be minimum, when the angle between E and ΔA is (a) 90° (b) zero (c) 45° (d) 60°	a
2. Which of the following cannot be the unit of electric intensity? (a) N/C (b) V/m (c) J/C-m (d) J/C	d
3. Electron volt is the unit of (a) Energy (b) Force (c) Potential difference (d) current	a
4. One joule per coulomb is equal to (a) Ampere (b) Volt (c) Farad (d) Tesla	b
5. The force per unit charge is known as: (a) Electric flux (b) Electric field intensity (c) Electric potential (d) Electric current	b
6. The Electric flux through a closed surface depends upon the (a) Magnitude of the charge enclosed by the surface. (b) Position of the charge enclosed by the surface. (c) The shape of the surface. (d) None of the above options.	a
7. In an electric field, the negative space rate of variation of potential at a point is equal to _____ at a point. (a) Potential gradient (b) Absolute potential (c) Electric intensity (d) Electric potential.	a
8. Decreasing the separation of two positive charges by one-half will cause the force of repulsion to change by: (a) 4 times (b) 2 times (c) $\frac{1}{2}$ times (d) $\frac{1}{4}$ times	a
9. The relation $-\frac{\Delta v}{\Delta r}$ represents (a) Gauss's law (b) Electric flux (c) Electric intensity (d) Potential difference	c
10. The introduction of a dielectric between the oppositely charged plate cause the intensity (a) To Increase (b) To decrease (c) To remain constant (d) To increase & Decrease	b
11. One electron volt is equal to (a) 1.6021×10^{-19} J (b) 9.11×10^{-27} J (c) 9.11×10^{-31} J (d) 6.67×10^{-11} J	a
12. What is an electric dipole moment? (a) The separation between two positive charges (b) The separation between two negative charges (c) The product of charge and distance between positive and negative charges (d) The product of charge and distance between like charges	c
13. What is the SI unit of electric dipole moment? (a) Coulomb (C)	d

- (b) Volt (V)
- (c) Ampere (A)
- (d) Coulomb-meter (C·m)

14. What happens to the electric dipole moment of an electric dipole when the magnitude of the charges increases?

- (a) The dipole moment decreases.
- (b) The dipole moment remains unchanged.
- (c) The dipole moment increases.
- (d) The dipole moment becomes zero.

15. What is the direction of the electric dipole moment?

- (a) From negative to positive charge
- (b) From positive to negative charge
- (c) Along the direction of electric field lines
- (d) Perpendicular to the electric field lines

16. What is the unit of electric flux?

- (a) V/m
- (b) N/m
- (c) V·m
- (d) N/Coulomb

17. The force of attraction between two unlike point charges is F , magnitude of force if distance between the point charges gets tripled will be.....

- (a) $9F$
- (b) $3F$
- (c) $(1/9)F$
- (d) $(1/3)F$

18. A charge of 6 C flows through the copper wire for 2 s connected across cell of potential difference 25 V, the work done is

- (a) 100 J
- (b) 150 J
- (c) 200 J
- (d) 250 J

19. The electric field inside a spherical surface distribution of charge is

- (a) Infinite
- (b) zero
- (c) KQ/r^2
- (d) KQ/r

at a point "p" placed on its axial line
ANSWER: See lecture(Electric Dipole)

Chapter 8

Chapter Formulae & Numericals

Coulomb's Law	$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$
Electric Field Intensity of a point Charge	$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
Electric Dipole	$\vec{P} = q \times d$
Electric Field at a point due to Dipole	$E = \frac{1}{4\pi\epsilon_0} \frac{P}{y^3}$
Electric Flux	$\phi_e = E \cdot A = EA \cos \theta$
Electric Flux through a sphere enclosing a charge	$\phi = \frac{q}{\epsilon_0}$
Potential Difference	$V = \frac{W}{q_0}$
Electric Potential of a Point Charge	$V_2 - V_1 = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$
Relation Between Electric Field and Electric Potential	$E = -\frac{\Delta V}{\Delta r}$